

MongoDB Multi-Collection Assignment 2

```
## Database: `library_db`
```

```
Collections:
```

```
    * `members`
```

```
    * `books`
```

```
    * `borrowings`
```

```
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```

```
## 1. Create Members Collection
```

```
```javascript
```

```
use("library_db")
```

```
db.members.insertMany([
```

```
 { member_id: 1, name: "Aarav", city: "Hyderabad", membership_type: "Gold" },
```

```
 { member_id: 2, name: "Priya", city: "Bangalore", membership_type: "Silver" },
```

```
 { member_id: 3, name: "Rahul", city: "Mumbai", membership_type: "Gold" },
```

```
 { member_id: 4, name: "Sneha", city: "Delhi", membership_type: "Silver" },
```

```
 { member_id: 5, name: "Kiran", city: "Hyderabad", membership_type: "Gold" }
```

```
])
```

```

```

```
2. Create Books Collection
```

```
```javascript
```

```
db.books.insertMany([
```

```
  { book_id: 101, title: "MongoDB Basics", category: "Database", author: "John Smith",  
    price: 500 },
```

```
  { book_id: 102, title: "Python Fundamentals", category: "Programming", author: "Alice  
    Brown", price: 650 },
```

```

    { book_id: 103, title: "Data Engineering Intro", category: "Data", author: "Mark Lee", price:
800 },

    { book_id: 104, title: "SQL for Beginners", category: "Database", author: "David Miller",
price: 450 },

    { book_id: 105, title: "Machine Learning Start", category: "AI", author: "Sara Khan", price:
900 }

])
'''

```

3. Create Borrowings Collection

```

```javascript
db.borrowings.insertMany([
 { borrow_id: 1001, member_id: 1, book_id: 101, days_borrowed: 5, borrow_date: "2026-
03-01" },

 { borrow_id: 1002, member_id: 2, book_id: 102, days_borrowed: 3, borrow_date: "2026-
03-02" },

 { borrow_id: 1003, member_id: 1, book_id: 103, days_borrowed: 7, borrow_date: "2026-
03-03" },

 { borrow_id: 1004, member_id: 3, book_id: 104, days_borrowed: 4, borrow_date: "2026-
03-05" },

 { borrow_id: 1005, member_id: 5, book_id: 105, days_borrowed: 10, borrow_date: "2026-
03-07" },

 { borrow_id: 1006, member_id: 5, book_id: 101, days_borrowed: 2, borrow_date: "2026-
03-08" }

])
'''

```

## Questions

### Basic Queries

1. Display all members.

**db.members.find()**

2. Display all books.

**db.books.find()**

3. Display all borrowings.

**db.borrowings.find()**

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### Filtering

4. Show members from **Hyderabad**.

**db.members.find({ city: "Hyderabad" })**

5. Show books in the **Database** category.

**db.books.find({ category: "Database" })**

6. Show books whose **price** is greater than 600.

**db.books.find({ price: { \$gt: 600 } })**

7. Show borrowings where **days\_borrowed** is greater than 5.

**db.borrowings.find({ days\_borrowed: { \$gt: 5 } })**

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## Sorting

8. Display books sorted by **price descending**.

```
db.books.find().sort({ price: -1 })
```

9. Display members sorted by **name ascending**.

```
db.members.find().sort({ name: 1 })
```

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## Counting

10. Count the total number of members.

```
db.members.countDocuments({})
```

11. Count the total number of books.

```
db.books.countDocuments({})
```

12. Count how many books belong to the **Database** category.

```
db.books.countDocuments({ category: "Database" })
```

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## Aggregation

13. Find the **average price of all books**.

```
db.books.aggregate([
 { $group: { _id: null, avgPrice: { $avg: "$price" } } }
])
```

14. Find the **maximum book price**.

```
db.books.aggregate([
 { $group: { _id: null, maxPrice: { $max: "$price" } } }
])
```

15. Find the **\*\*minimum book price\*\***.

```
db.books.aggregate([
 { $group: { _id: null, minPrice: { $min: "$price" } } }
])
```

16. Find the **\*\*total days borrowed per member\*\***.

```
db.borrowings.aggregate([
 { $group: { _id: "$member_id", totalDays: { $sum: "$days_borrowed" } } }
])
```

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## Multi-Collection Queries using `\$lookup`

17. Display borrowings along with **\*\*member details\*\***.

```
db.borrowings.aggregate([
 {
 $lookup: {
 from: "members",
 localField: "member_id",
 foreignField: "member_id",
 as: "member_details"
 }
 }
])
```

18. Display borrowings along with **\*\*book details\*\***.

```
db.borrowings.aggregate([
 {
 $lookup: {
 from: "books",
 localField: "book_id",
 foreignField: "book_id",
 as: "book_details"
 }
 }
])
```

19. Display **\*\*member name and book title\*\*** for each borrowing.

```
db.borrowings.aggregate([
 { $lookup: { from: "members", localField: "member_id", foreignField: "member_id",
as: "m" } },
 { $lookup: { from: "books", localField: "book_id", foreignField: "book_id", as: "b" }
},
 { $unwind: "$m" },
 { $unwind: "$b" },
 { $project: { _id: 0, member_name: "$m.name", book_title: "$b.title" } }
])
```

20. Display **\*\*book title and total times it was borrowed\*\***.

```
db.borrowings.aggregate([
 { $group: { _id: "$book_id", count: { $sum: 1 } } },
 { $lookup: { from: "books", localField: "_id", foreignField: "book_id", as:
"book_info" } },
 { $unwind: "$book_info" },
 { $project: { _id: 0, book_title: "$book_info.title", total_borrows: "$count" } }
])
```

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## Advanced Aggregation

21. Find the **\*\*total number of books borrowed by each member\*\***.

```
db.borrowings.aggregate([
 { $group: { _id: "$member_id", totalBooks: { $sum: 1 } } }
])
```

22. Find the **\*\*most borrowed book\*\***.

```
db.borrowings.aggregate([
 { $group: { _id: "$book_id", count: { $sum: 1 } } },
 { $sort: { count: -1 } },
 { $limit: 1 },
 { $lookup: { from: "books", localField: "_id", foreignField: "book_id", as: "details" } },
 { $unwind: "$details" },
 { $project: { _id: 0, title: "$details.title", count: 1 } }
])
```

23. Find the **\*\*total borrowing count by category\*\***.

```
db.borrowings.aggregate([
 { $lookup: { from: "books", localField: "book_id", foreignField: "book_id", as: "book" } },
 { $unwind: "$book" },
 { $group: { _id: "$book.category", count: { $sum: 1 } } }
])
```

24. Find members who borrowed **\*\*more than one book\*\***.

```
db.borrowings.aggregate([
 { $group: { _id: "$member_id", count: { $sum: 1 } } },
 { $match: { count: { $gt: 1 } } }
])
```

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## Final Challenge

25. Display:

- \* member name
- \* city
- \* total books borrowed

Sort by **\*\*highest books borrowed first\*\***.

```
db.members.aggregate([
 {
 $lookup: {
 from: "borrowings",
 localField: "member_id",
 foreignField: "member_id",
 as: "history"
 }
 },
 {
 $project: {
 _id: 0,
 name: 1,
 city: 1,
 total_books_borrowed: { $size: "$history" }
 }
 },
 { $sort: { total_books_borrowed: -1 } }
])
```

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